

Unit 5:

Css:

- Css stands for Cascading Style Sheet.
- used for styling the html document.
- used controlling the appearance of html elements
- It describes how elements should be displayed
- css id defined inside the style tag
- like as other html tag css has also opening and closing tag i.e <style> and </style>
- css are of three types
 - inline
 - internal
 - external
- the extension of css file is .css

The general syntax for defining the css is:

```
<!Doctype html>
<head>
<title>
css
</title>
</head>

<style>
-----
-----
-----
</style>
<body>
-----
</body>
</html>
```

Css includes the components like selector, declaration, property/attributes, value:

selector: It is used to determine in which tag the styling is done

declaration: It represent the content inside the style

property: This represent the what styling actually adding on that tag

value: The value represent the attributes value

Example:

```
<!Doctype html>
<head>
<title>
Css
</title>
<style>
p{
color: red;
}
</style>
</head>
<body>
<p> This is paragraph tag</p>
</body>
</html>
```

Here, above there is a <p> tag, styling of <p> tag is done inside the <style>

CSS selectors:

CSS elements selects the tag/elements that you want to style

The common selectors are

- elements selector- It is simply represented by tag name (p, div, h,)
- id selector- It is represented by #tag_name (#profile1, #group)
- class selector- It is represented by .class_name (.container, .box)

Example:

Element selector:

```
p {  
  text-align: center;  
  color: red;  
}
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
p {  
  text-align: center;  
  color: red;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p>Every paragraph 1.</p>
```

```
<p>paragraph 2</p>
```

```
</body>
```

```
</html>
```

Example:

id selector:

```
#id1 {  
  text-align: center;  
  color: red;  
}
```

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
#id1 {  
  text-align: center;  
  color: red;  
}  
</style>  
</head>  
<body>  
<p id="id1">id selector1</p>  
<p>id selector</p>  
</body>  
</html>
```

Example:
class selector:

```
.center {  
  text-align: center;  
  color: red;  
}
```

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
.center {  
  text-align: center;  
  color: red;  
}  
</style>  
</head>  
<body>  
  
<h1 class="center">class selector</h1>  
<p class="center">next class selector</p>  
  
</body>  
</html>
```

Types of Style sheet:

There are three types of style sheet:

- inline css
- internal css
- external css

Inline css:

-Inline CSS involves applying styles directly to individual HTML elements using the style attribute.

-This method allows for specific styling of elements within the HTML document, overriding any external or internal styles.

```

<!Doctype html>
<head>
<title>
Using online css
</title>
</head>
<body>
<p style="color:#009900;
        font-size:50px;
        font-style:italic;
        text-align:center;">
    Inline CSS
</p>
</body>
</html>

```

internal css:

- Internal css is also known as embedded css.
- Internal css is defined within the html document.
- css is define in the head section.

The CSS is embedded within the <style> tag inside the head section of the HTML file.

```

<!DOCTYPE html>
<html>
<head>
    <style>
        p{
            color: #009900;
            font-size: 50px;
            font-weight: bold;
            font-style: bold;
        }
    </style>
</head>
<body>
    <p>Internal CSS</p>
</body>
</html>

```

external css:

- External CSS contains separate CSS files.
- External css file that only style properties with the help of tag attributes
- CSS property is written in a separate file with a .css extension
- Css should be linked to the HTML document using a link tag.
- for each element, style can be set only once and will be applied across web pages.

external css is linked in the html document by link elements:

```
<link rel="stylesheet" href="style.css">
```

html document:

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<p>This is paragraph tag</p>
</body>
</html>
```

css document:

```
p{
    color: red;
}
```

Advantages of css:

1. Separation of Content from Design
2. Greater Design Flexibility
3. CSS style Consistency
4. Responsive Design
5. Customizable Styling

Disadvantages:

1. Cross-Browser Compatibility Issues
2. Complexity in Large Projects
3. Dependency on External Files
4. Overhead with Inline Styles
3. No Built-in Security

The various types of styles and attributes are as follows:

- Backgrounds
- Colors
- Fonts
- Text
- Margin
- Padding
- Border
- List

Background:

It is used to set the style the background of html element

The property of background are as follows:

- background-color
- background-image
- background-position
- background-size
- background-repeat

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body {
```

```
    background-color: blue;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>background</h1>
```

```
<p>
```

```
color added
```

```
</p>
```



```
</body>
</html>
```

Font:

- It makes easy to read.
- font adds value to the text.
- correct color and text size increase the appearance of text

The properties of font:

- font-family
- font-size
- font-weight

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
h1{
font-family: Arial
color: red;
}
</style>
</head>
<body>
```

```
<h1>Font</h1>
```

```
<p>
color added
</p>
</body>
</html>
```

Text:

- To enhance the text appearance

Text properties are:

- text-color:
- text-size
- text-transform
- text-alignment
- text-decoration

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
h1{
text-transform: upper;
color: red;
}
</style>
</head>
<body>

<h1>Text </h1>

<p>
color added
</p>
</body>
</html>
```

margin:

- Margins are used to create space around elements, outside of any defined borders.
- The attributes of margin
 - margin-top
 - margin-left
 - margin-right
 - margin-bottom

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  margin: 70px;
  border: 1px solid #4CAF50;
}
</style>
</head>
<body>
<div>Margin</div>
</body>
</html>
```

margin:

-Padding is used to create space around an element's content, inside of any defined borders.

The attributes of margin

- margin-top
- margin-bottom

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  padding: 70px;
  border: 1px solid #4CAF50;
}
</style>
</head>
<body>

<h2>CSS Padding</h2>
<div>This element has a padding of 70px.</div>

</body>
</html>
```

Border:

The CSS border properties allow you to specify the style, width, and color of an element's border.

the properties of border are as follows:

- border-width
- border-color
- border-style
- border-radius

- List:

- Set different list item markers for ordered lists
- Set different list item markers for unordered lists
- Set an image as the list item marker
- Add background colors to lists and list items

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
ul.ul1 {
  list-style-type: circle;
}

ul.ul2 {
  list-style-type: square;
}

ol.ul3{
  list-style-type: upper-roman;
}

</style>
</head>
<body>
```

<h2>The list-style-type Property</h2>

<p> unordered lists:</p>

```
<ul class="ul1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Coca Cola</li>
</ul>
```

```
<ul class="ul2">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Coca Cola</li>
</ul>
```

<p>of ordered lists:</p>

```
<ol class="ul3">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Coca Cola</li>
</ol>
```

```
</body>
</html>
```

- Table:

It is used to styling the table in the html document

The property of tables are

- border
- alignment
- size

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
table, th, td {
  border: 1px solid;
}
</style>
</head>
<body>
```

```
<h2>Add a border to a table:</h2>
```

```
<table>
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
  </tr>
  <tr>
    <td>Ram Sharma</td>
    <td>Jeevan K C</td>
  </tr>
  <tr>
    <td>Hari Rana</td>
    <td>Seema Chand</td>
  </tr>
</table>
</body>
</html>
```

Div tag:

-The <div> tag defines a division or a section in an HTML document.

-The <div> tag is used as a container for HTML elements - which is then styled with CSS or manipulated with JavaScript.

-The <div> tag is easily styled by using the class or id attribute.

-Any sort of content can be put inside the <div> tag

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
div {
```

```
    display: block;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div>box 1</div>
```

```
<div> box2</div>
```

```
</body>
```

```
</html>
```

Responsive Web Design:

- Responsive web design is about creating web pages that look good on all devices.
- A responsive web design will automatically adjust for different screen sizes and viewports.
- Responsive Web Design is about using HTML and CSS to automatically resize, hide, shrink, or enlarge, a website, to make it look good on all devices (desktops, tablets, and phones):
to create responsive add this on head section of the html document
<meta name="viewport" content="width=device-width, initial-scale=1.0">

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
<h2> Viewport</h2>
<p>Viewport added</p>
</body>
</html>
```

Example:

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<style>
* {
  box-sizing: border-box;
}
.menu {
  float: left;
  width: 20%;
}
.menuitem {
  padding: 8px;
  margin-top: 7px;
  border-bottom: 1px solid #f1f1f1;
}
```



```
.main {
  float: left;
  width: 60%;
  padding: 0 20px;
  overflow: hidden;
}
.right {
  background-color: lightblue;
  float: left;
  width: 20%;
  padding: 10px 15px;
  margin-top: 7px;
}
```

```
@media only screen and (max-width:800px) {
  /* For tablets: */
  .main {
    width: 80%;
    padding: 0;
  }
  .right {
    width: 100%;
  }
}
```

```
@media only screen and (max-width:500px) {
  /* For mobile phones: */
  .menu, .main, .right {
    width: 100%;
  }
}
```

</style>

</head>

<body style="font-family:Verdana;">

```
<div style="background-color:#f1f1f1;padding:15px;">
  <h1>Cinque Terre</h1>
  <h3>Resize the browser window</h3>
</div>
```

```
<div style="overflow:auto">
  <div class="menu">
    <div class="menuitem">The Walk</div>
    <div class="menuitem">Transport</div>
    <div class="menuitem">History</div>
```

```
<div class="menuitem">Gallery</div>
</div>
```

```
<div class="main">
  <h2>The Walk</h2>
  <p>The walk from Monterosso to Riomaggiore will take you approximately two hours, give or
take an hour depending on the weather conditions and your physical shape.</p>
  
</div>
```

```
<div class="right">
  <h2>What?</h2>
  <p>Cinque Terre comprises five villages: Monterosso, Vernazza, Corniglia, Manarola, and
Riomaggiore.</p>
  <h2>Where?</h2>
  <p>On the northwest cost of the Italian Riviera, north of the city La Spezia.</p>
  <h2>Price?</h2>
  <p>The Walk is free!</p>
</div>
</div>
```

```
<div
style="background-color:#f1f1f1;text-align:center;padding:10px;margin-top:7px;font-size:12px;">
This web page is a part of a demonstration of fluid web design made by w3schools.com. Resize
the browser window to see the content respond to the resizing.</div>
```

```
</body>
</html>
```


